

The Jag2/Notch1 signaling axis promotes sebaceous gland differentiation and controls progenitor proliferation

Reviewed Preprint

v2 • September 30, 2024

Revised by authors

Reviewed Preprint

v1 • July 9, 2024

Syeda Nayab Fatima Abidi , Sara Chan, Kerstin Seidel, Daniel Lafkas, Louis Vermeulen, Frank Peale, Christian W Siebel

Department of Discovery Oncology, Genentech, South San Francisco, USA • Department of Research Pathology, Genentech, South San Francisco, USA

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

The sebaceous gland (SG) is a vital appendage of the epidermis, and its normal homeostasis and function is crucial for effective maintenance of the skin barrier. Notch signaling is a well-known regulator of epidermal differentiation, and has also been shown to be involved in postnatal maintenance of SGs. However, the precise role of Notch signaling in regulating SG differentiation in the adult homeostatic skin remains unclear. While there is evidence to suggest that Notch1 is the primary Notch receptor involved in regulating the differentiation process, the ligand remains unknown. Using monoclonal therapeutic antibodies designed to specifically inhibit of each of the Notch ligands or receptors, we have identified the Jag2/Notch1 signaling axis as the primary regulator of sebocyte differentiation in homeostatic skin. Mature sebocytes are lost upon specific inhibition of the Jag2 ligand or Notch1 receptor, resulting in the accumulation of proliferative stem/progenitor cells in the SG. Strikingly, this phenotype is reversible, as these stem/progenitor cells re-enter differentiation when the inhibition of Notch activity is lifted. Thus, Notch activity promotes correct sebocyte differentiation, and is required to restrict progenitor proliferation.

eLife assessment

This work aimed at deconstructing how sebaceous gland differentiation is controlled in adult skin. Using monoclonal antibodies designed to inhibit specific Notch ligands or receptors, the authors present **convincing** evidence that the Jag2/Notch1 signaling axis is a crucial regulator of sebocyte progenitor proliferation and sebocyte differentiation. The **valuable** findings presented here contribute to the growing evidence that Notch signaling is not only key during the development of the skin and its appendages but also regulates cell fate in adult homeostatic tissues. From a translational perspective, it is intriguing that the effect of Jag2 or Notch1 inhibition, which leads to the accumulation of proliferative stem/progenitor cells in the sebaceous gland and prevents sebocyte differentiation, is reversible.

<https://doi.org/10.7554/eLife.98747.2.sa4>

Introduction

The skin is a vital organ that acts as a protective barrier against the external environment, and safeguards against fluid loss. An important component of this barrier function is the presence of a complex mixture of oils, known as sebum, which is produced by the sebaceous glands (SGs). SGs are part of the epidermis and are typically associated with the hair follicle. These acinar structures have two cell types: basal stem or progenitor cells which encase the differentiated sebocytes (**Figure 1a**). Sebocyte differentiation begins at the proximal tip of the of the SG, with maturing sebocytes moving upwards, enlarging, accumulating lipids, and ultimately undergoing a highly regulated and specialized form of cell death in which they release their lipid contents into the sebaceous duct (**Figure 1b**) (Kretzschmar et al., 2014; Schneider and Paus, 2010). This process requires the constant turnover of sebocytes, which occurs over a period of 7-14 days in mice (Jung et al., 2015). Both over- and underproduction of sebum have been linked to various skin disorders including acne or dry skin (Al-Zaid et al., 2011; Binczek et al., 2007; Karnik et al., 2009; Lovász et al., 2017; Rittié et al., 2016; Seiffert et al., 2007; Shi et al., 2015; Smith and Thiboutot, 2008; Stenn et al., 1999), and rare sebaceous carcinomas constitute aggressive tumors leading to high mortality (Buitrago and Joseph, 2008; Nelson et al., 1995), thus SG number and function have to be tightly regulated for proper skin function.

The Notch signaling pathway is one of the most studied regulators of cell fate decisions, and is known to be widely involved in epidermal differentiation. The pathway consists of multiple ligands and receptors that typically form a signaling axis in pairs. Notch signaling can regulate cell fate by either inducing or inhibiting differentiation, or by making binary cell fate decisions (Wilson and Radtke, 2006). Classically, these functions of Notch signaling have been studied during development, but increasing evidence suggests that the Notch pathway is also involved in regulating cell fate and cell states in adult homeostatic tissues (Ables et al., 2011; Lafkas et al., 2015; Mosteiro et al., 2023; Sato et al., 2012; Siebel and Lendahl, 2017).

While it is known that Notch signaling is not required for embryonic development of the epidermis, it is essential for the postnatal maintenance of the hair follicles and the SGs (Watt et al., 2008). However, the precise role of Notch signaling in adult sebocyte differentiation has not been comprehensively investigated, with most studies examining irreversible deletions of the Notch pathway components in the embryonic ectodermal lineages. While these studies report SG defects, it remains unclear whether these defects are due to a direct effect on the SGs, or whether they are a consequence of general skin defects also observed in these models. For example, SGs are absent in mice with embryonic pan-Notch deletions such as *RBPJ* (Blanpain et al., 2006), gamma-secretase, *Notch1Notch2*, and *Notch1Notch2Notch3*, and are severely reduced in *Notch1* and *Notch1Notch3* embryonically-deleted skin (Pan et al., 2004), while deletion of *Notch2* alone does not affect the SG (Pan et al., 2004). Consistent with the constitutive deletions, loss of *RBPJ* in adult SGs also results in missing sebocytes, while loss of *Notch1* in the adult SGs results in miniaturized lobes that still contain some differentiated sebocytes (Veniaminova et al., 2019). Interestingly, activation of *Notch1* in the adult skin results in enlarged SGs (Estrach et al., 2006). Conversely, for the Notch pathway ligands, embryonic and adult deletion of *Jag1* (Estrach et al., 2006), and embryonic deletion of *Dll1* (Estrach et al., 2008) results in normal SG morphology. Collectively, these data suggest that Notch1 is the dominant Notch receptor involved in regulating sebocyte differentiation, however, it remains unclear which ligand is required. In our previous work, we observed that systemic inhibition of Jag2 using monoclonal therapeutic antibodies resulted in SG defects in adult mice (Lafkas et al., 2015). Given that our therapeutic antibodies selectively, potently, and transiently inhibit each of the distinct Notch receptors and ligands (Tran et al., 2013; Wu et al., 2010; Yu et al., 2020), they constitute ideal tools to dissect the contribution of each of these pathway members to sebocyte differentiation in adult homeostatic skin. Leveraging the use of these antibodies, we demonstrate that specific inhibition of the Jag2

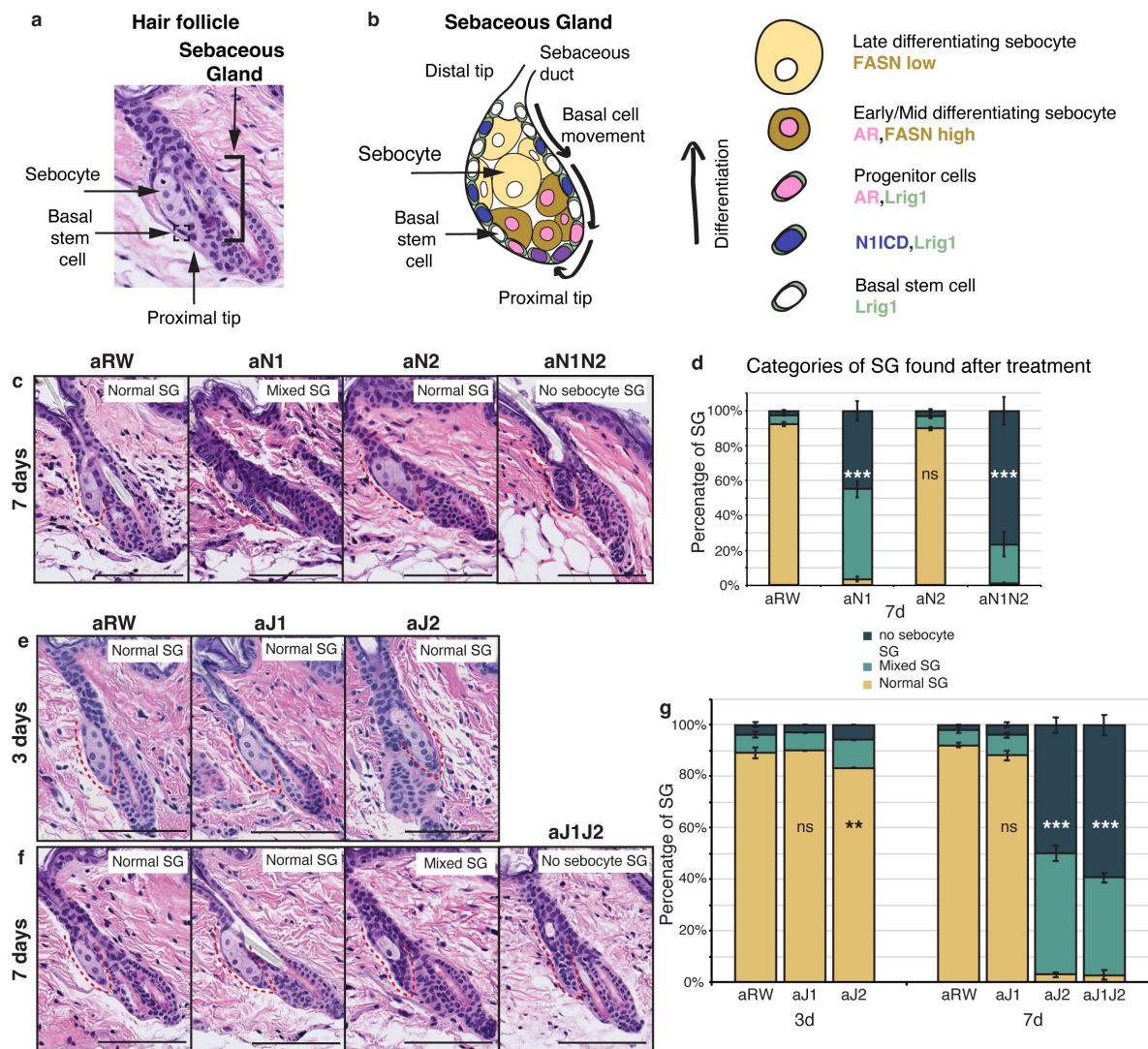


Figure 1.

Jag2 is the dominant ligand involved in regulating sebocyte differentiation.

(a) A representative image of the homeostatic hair follicle, including its associated sebaceous gland (SG). (b) A detailed schematic of the SG showing the outer basal stem cell layer encasing the differentiated sebocytes. The schematic also shows the gene expression in different regions of the SG. (c) Representative H&E images of SGs from mice ($n = 5$ each) treated with aRW, aN1, aN2 and aN1N2, 7 days post treatment. (d) Quantification of type of SG found after each treatment. The SGs were divided into three categories: normal SGs containing characteristic number of sebocytes (normal SG), SGs containing a mix of sebocytes and basal-like cells (mixed SG), and SGs containing no sebocytes and only basal-like cells (no sebocyte SG). For (c), (e) and (f), each SG panel is labeled with where the SG is normal, mixed or no sebocyte SG. p-values: aN1 = $2.02\text{E-}123$, aN2 = 0.426 , aN1N2 = $3.85\text{E-}114$. Total n of SGs quantified per treatment: aRW = 425, aN1 = 298, aN2 = 343, aN1N2 = 217. (e) Representative H&E images of SGs from mice ($n = 5$ each) treated with aRW, aJ1 and aJ2, 3 days post treatment. (f) Representative H&E images of SGs from mice ($n = 5$ each) treated with aRW, aJ1, aJ2 and aJ1J2, 7 days post treatment. (g) Quantification of type of SG found after each treatment. 3 days: p-values: aJ1 = 0.409 , aJ2 = 0.004 . 7 days: p-values: aJ1 = 0.057 , aJ2 = $7.61\text{E-}151$, aJ1J2 = $6.23\text{E-}169$. Total n of SGs quantified per treatment: 3 days, aRW = 851, aJ1 = 944, aJ2 = 760, 7 days, aRW = 491, aJ1 = 499, aJ2 = 388, aJ1J2 = 461. Chi-square test used for statistical analysis. All treatments compared against aRW. Error bars represent SEM. Scale bars are $100\text{ }\mu\text{m}$.

ligand or Notch1 receptor both result in the loss of mature sebocytes in the SG, establishing the Jag2/Notch1 signaling axis as a crucial regulator of sebocyte differentiation in adult homeostatic skin. The loss of mature sebocytes in the SG is concomitant with an accumulation of cells with a basal phenotype, forming epithelial remnants in the SG. Cells in these epithelial remnants are actively proliferating and express stem/progenitor markers indicating that sebocyte differentiation is halted, while stem/progenitor numbers are increased. Importantly, this phenotype is reversible, as these epithelial cells re-enter differentiation with the return of Notch activity. Thus, Notch activity is required in sebocyte stem/progenitor cells for their proper differentiation, and its inhibition locks these cells in a reversible progenitor state.

Results

Jag2 is the dominant Notch signaling ligand involved in regulating sebocyte differentiation

To investigate the role of Notch signaling in the homeostatic skin, we treated 8-week old mice with a single dose of the various antagonizing antibodies (anti-Notch1, anti-Notch2, anti-Jag1, anti-Jag2 and the isotype control antibody anti-Ragweed) alone or in combination. We then examined the dorsal, resting phase (telogen) skin at 3, 7 and 14 days post treatment. We first confirmed that our antibodies could reproduce the requirement of Notch1 in regulating sebocyte differentiation. To this end, we treated mice with Notch1 (aN1) and Notch2 (aN2) blocking antibodies and examined the SGs at 7 days post treatment. Loss of mature sebocytes was observed specifically after aN1 treatment, while the aN2 treated SGs showed normal morphology (**Figure 1c,d**). However, the combined treatment of aN1N2 had a more pronounced effect on SG morphology as compared to aN1 alone (**Figure 1c,d**), indicating that Notch2 also contributes to regulating sebocyte differentiation, possibly as a compensatory mechanism after inhibition of the dominant Notch1 receptor.

Next, we investigated which ligand formed the signaling pair with Notch1. Loss of mature sebocytes was observed specifically after treatment with the Jag2 blocking antibody (aJ2), but not after treatment with the Jag1 blocking antibody (aJ1) at 7 days post treatment (**Figure 1e,f**). For aJ2 treatment, the loss of sebocytes began at 3 days post treatment, but affected only a small proportion of the SGs at this timepoint (**Figure 1g**). However, by 7 days post treatment, most SGs had either completely lost all sebocytes, or had only some sebocytes remaining (**Figure 1g**). The loss-of-sebocyte phenotype was most pronounced with a combined treatment of Jag1/Jag2 blocking antibodies (aJ1J2) (**Figure 1g**), suggesting that while Jag2 is the primary Notch ligand involved in regulating sebocyte differentiation, Jag1 also plays a minor role in this process, but is unable to produce a phenotype on its own. In conclusion, these data indicate that the Jag2-Notch1 signaling axis is the dominant Notch ligand-receptor pair required for sebocyte differentiation in the adult skin.

We also examined the percentage of SGs consisting of bursting sebocytes releasing sebum as a proxy of a functional sebaceous duct. There were no significant differences between treatments (**Figure S1a,b**), hinting at a functionally intact sebaceous duct.

Notch is active in the sebaceous gland stem cells

We next characterized the expression of the relevant receptor and ligand in the SG to develop a spatial map of Notch activity in this tissue. Both the Notch receptors and their ligands are transmembrane proteins that interact with each other on neighboring cells to activate the pathway. To examine this interaction, we performed a triple stain for the cleaved (active) form of the Notch1 intracellular domain (ICD), and Notch1 and Jag2 in situ hybridization (ISH) probes at 3 days post antibody treatment. In the control treated mice, Notch1 ICD was observed in the basal

stem cell compartment of the SG, but not in the differentiating sebocytes (**Figure 2a-c** [↗](#), cell types based on morphology). We saw a similar pattern of N1ICD-positive cells for aJ1 treatment (**Figure 2d-f** [↗](#)). Notch1 signaling was not active in all basal cells, as only ~50% of them were positive for N1ICD after control and aJ1 treatment (**Figure 2g** [↗](#)). Consistent with previous studies, the majority of these N1ICD+ cells were present near the proximal tip of the SG (Figure S2a-c) (Veniaminova et al., 2019 [↗](#)), where the initial sebocyte differentiation has been proposed to occur (Kretzschmar et al., 2014 [↗](#)). However, after aJ2 treatment, Notch1 activity (ICD staining) was absent, or observed only at very low levels in the basal stem cells, while they still expressed Notch1 and Jag2 mRNA (**Figure 2h-j** [↗](#)). Strikingly, we noticed the expression of both Notch1 ISH and Jag2 ISH in the same cell on most N1ICD+ cells (**Figure 2k** [↗](#)). Interestingly, a majority of all basal cells (including N1ICD-cells), expressed both N1 and Jag2 mRNA (Figure S2d). Our data shows that the vast majority of SG basal stem cells express both the ligand and receptor, but only some basal stem cells experience active Notch signaling at any one point in time, as evidenced by the presence of N1ICD. This could be a technical limitation as the triple staining only captures a temporal snapshot of the basal stem cells, with the N1+J2 mRNA positive cells going on to express N1ICD later, or it could hint towards a more complex regulatory mechanism involved in activating Notch signaling.

Notch activity, as well as Notch1 and Jag2 mRNA were also observed in the interfollicular epidermis (IFE) and other cells of the hair follicle (Figure S2e,f). While there is a reduction in Notch-active cells in these other regions, it does not appear to significantly impact the rest of the skin. There were no significant differences in the width of the IFE or the adipocyte layer between treatments (Figure S2g,h). This together with the histological appearance (Figure S2i,j) of these regions suggests that proliferation and differentiation in these compartments remains unaffected.

Loss of Notch activity in the SG stem cells inhibits sebocyte differentiation

To further detail the effect of Notch inhibition on sebocyte differentiation, we examined the expression of mature sebocyte markers. Adipophilin (Adipo) is expressed in all mature sebocytes (Frances and Niemann, 2012 [↗](#); Ostler et al., 2010 [↗](#)) and Fatty acid synthase (FASN) is expressed in mid- and late-differentiating sebocytes, with FASN levels decreasing in the most mature sebocytes (Cottle et al., 2013 [↗](#)) (**Figure 1b** [↗](#)). Since N1ICD staining disappears at 3 days post antibody treatment, we examined the mature sebocyte markers at this timepoint. While N1ICD staining was specifically lost in the SG basal stem cells of aJ2 treated skin, mature sebocytes expressing FASN were still observed at this timepoint (**Figure 2j** [↗](#), **Figure 3a,b** [↗](#) and Figure S3a). All cells that express FASN also express adipophilin, but since FASN levels decrease with sebocyte maturity, we also examined and focused on adipophilin to mark all sebocytes (**Figure 3c,d** [↗](#) and Figure S3b-e). The number of cells that expressed adipophilin were not significantly different between treatments at 3 days post antibody treatment (**Figure 3h** [↗](#)). At 7 days post antibody treatment, control and aJ1 treated skin showed normal SG morphology and sebocyte marker expression (**Figure 3e** [↗](#) and Figure S3f,h,i), but both aJ2 and aJ1J2 SGs had lost sebocyte marker expression, and the SG was filled with cells with a basal phenotype (basal-like cells) (**Figure 3f** [↗](#) and Figure S3g,j,k). Consistently, the number of cells expressing adipophilin were significantly lower for aJ2 and aJ1J2 treatment (**Figure 3i** [↗](#)). Interestingly, we noticed that some of these affected SGs still contained a few mature sebocytes (**Figure 3f** [↗](#) and Figure S3g), which were found at the distal end near the sebaceous duct. The location of these remaining sebocytes suggests that existing mature sebocytes are not affected by the Notch blockade, and go through their normal differentiation process, eventually bursting and releasing the sebum (**Figure 3g** [↗](#)). Thus, we propose that Notch blockade inhibits differentiation at basal stem cell-level or in a sebocyte progenitor.

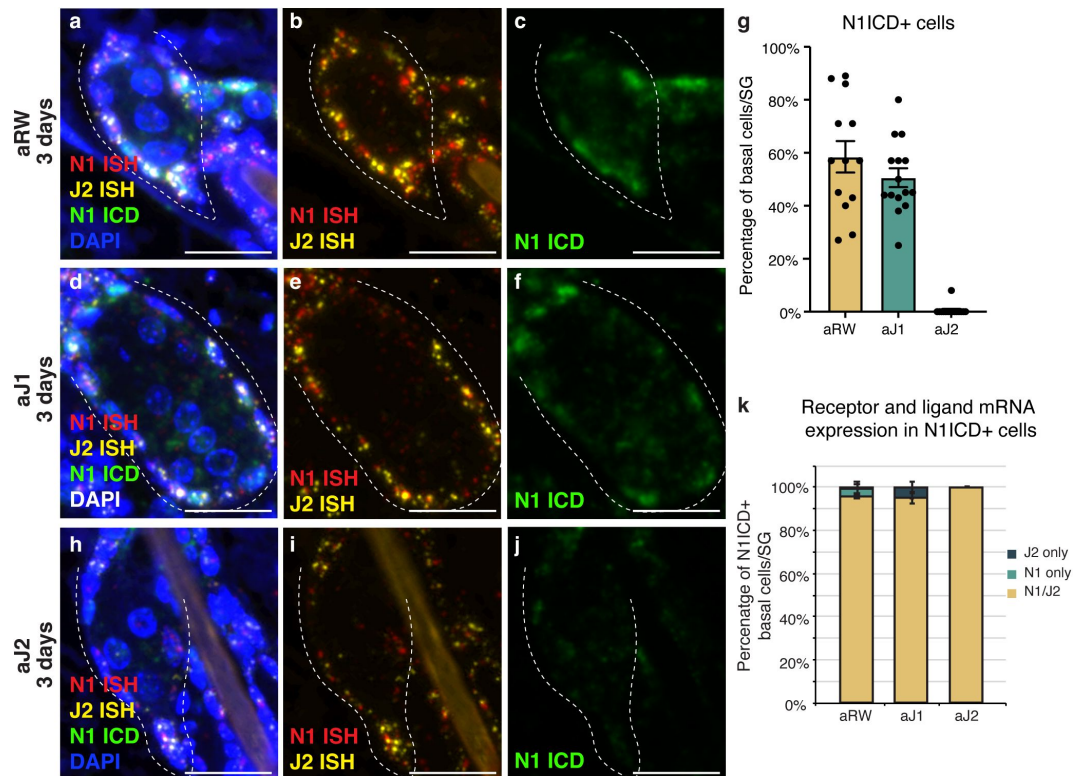


Figure 2.

Notch is active in the sebaceous gland stem cells.

(a-c) Representative triple stain images for N1ICD, Notch1 and Jag2 mRNA in SGs from mice ($n = 5$ each) treated with aRW, 3 days post treatment. (a) Four channels showing N1ICD, N1 ISH, J2 ISH and DAPI. (b) Two channels showing N1 ISH and J2 ISH. (c) One channel showing N1ICD. (d-f) Representative triple stain images for N1ICD, Notch1 and Jag2 mRNA in SGs from mice ($n = 5$ each) treated with aj1, 3 days post treatment. (d) Four channels showing N1ICD, N1 ISH, J2 ISH and DAPI. (e) Two channels showing N1 ISH and J2 ISH. (f) One channel showing N1ICD. (g) Quantification of the percentage of N1ICD positive (N1ICD+) basal stem cells in SGs from mice ($n = 5$ each) treated with aRW, aj1 and aj2, 3 days post treatment. Percentage was calculated by dividing the number N1ICD+ basal stem cells by total number of basal stem cells in each SG. (h-j) Representative triple stain images for N1ICD, Notch1 and Jag2 mRNA in SGs from mice ($n = 5$ each) treated with aj2, 3 days post treatment. (h) Four channels showing N1ICD, N1 ISH, J2 ISH and DAPI. (i) Two channels showing N1 ISH and J2 ISH. (j) One channel showing N1ICD. (k) Quantification of what percentage of the N1ICD+ basal stem cells express both N1 ISH and J2 ISH, only N1 ISH or only J2 ISH. (g and k) Total n of SGs quantified per treatment: aRW = 13, aj1 = 15, aj2 = 15. Error bars represent SEM. Scale bars are 25 μ m.

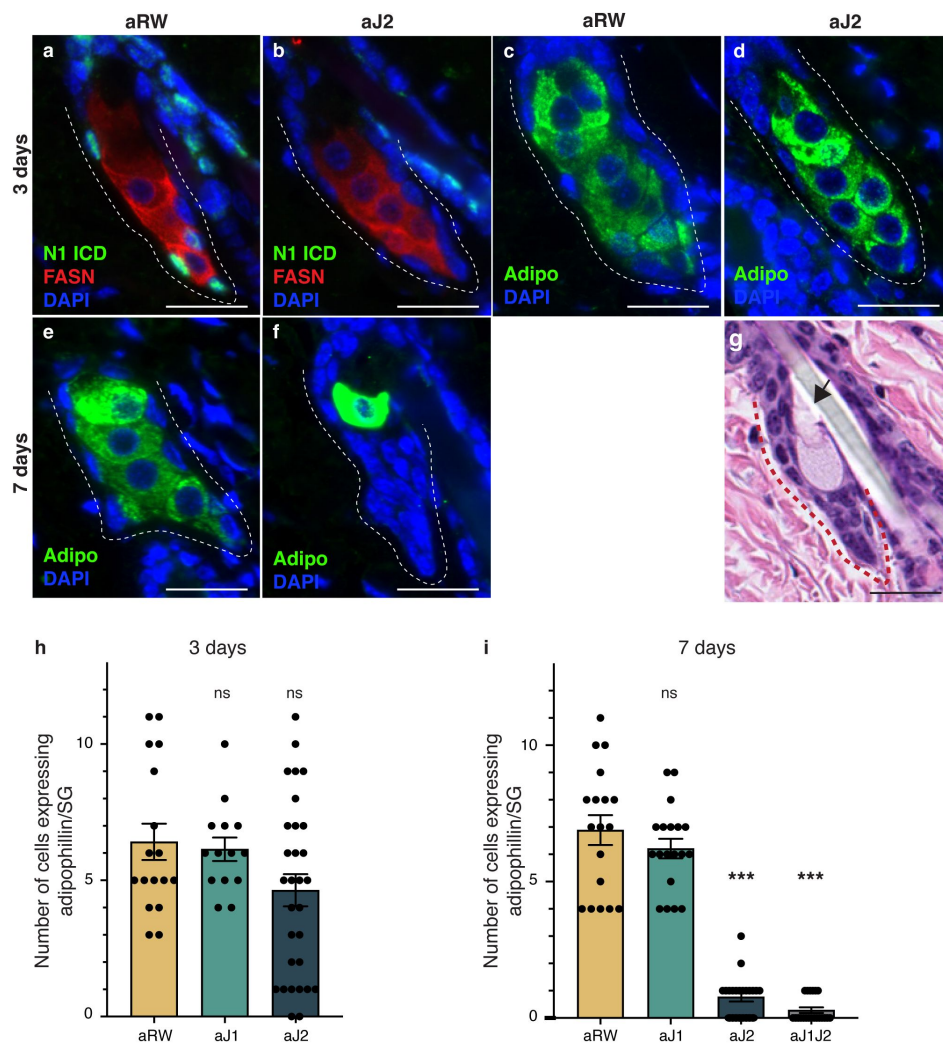


Figure 3.

Loss of Notch activity in the SG stem cells inhibits sebocyte differentiation.

(a,b) Representative co-stain images for N1ICD and FASN in SGs from mice ($n = 5$ each) treated with aRW (a) and aJ2 (b), 3 days post treatment. (c,d) Representative adipophillin staining in SGs from mice ($n = 5$ each) treated with aRW (c) and aJ2 (d), 3 days post treatment. (e,f) Representative adipophillin staining in SGs from mice ($n = 5$ each) treated with aRW (e) and aJ2 (f), 7 days post treatment. (g) A representative H&E image of an SG from mice ($n = 5$ each) treated with aJ2, 7 days post treatment. Arrow points to a bursting sebocyte releasing sebum into the sebaceous duct. (h) Quantification of the number of cells expressing adipophillin in each SG, 3 days post treatment with aRW, aJ1 and aJ2. p-values: aJ1 = 0.749, aJ2 = 0.062. Total n of SGs quantified per treatment: aRW = 17, aJ1 = 14, aJ2 = 30. (i) Quantification of the number of cells expressing adipophillin in each SG, 7 days post treatment with aRW, aJ1, aJ2 and aJ1J2. p-values: aJ1 = 0.301, aJ2 = 4.13E-14, aJ1J2 = 4.06E-14. Total n of SGs quantified per treatment: aRW = 18, aJ1 = 19, aJ2 = 22, aJ1J2 = 21. Student's T-test used for statistical analysis. All treatments compared against aRW. Error bars represent SEM. Scale bars are 25 μ m.

Notch activity in the SG stem cells is required to prevent unregulated progenitor proliferation

To determine whether the epithelial cells that filled the affected SGs at 7 days post treatment were stem/progenitor cells, we examined them for stem and early differentiation markers. Lrig1 positive cells form a distinct stem cell compartment that maintains the SG and the upper part of the hair follicle (Frances and Niemann, 2012 [DOI](#); Niemann and Horsley, 2012 [DOI](#); Page et al., 2013 [DOI](#)), allowing us to use Lrig1 as a marker for the basal stem cells of the SG, while Androgen Receptor (AR) can be used as an early marker of sebocyte differentiation (Cottle et al., 2013 [DOI](#)) (Figure 4a,b [DOI](#) and Figure S4a,b). We noticed that the AR-expressing cell population could be divided into two groups: a basal stem cell population that co-expressed Lrig1, but did not express FASN (arrow heads in Figure 4a,c [DOI](#)), and an early differentiating sebocyte population that expressed FASN, but did not express Lrig1 (arrows in Figure 4a,c [DOI](#)). We hypothesize that the Lrig1+/AR+ population is a progenitor cell population, in addition to the Lrig1+ stem cells, similar to the recently identified transitional basal cell population in the SG (Veniaminova et al., 2023 [DOI](#)). There were no significant differences in the number of Lrig1 positive stem cells per SG, or the AR-expressing progenitor population between treatments at 3 days post treatment (Figure 4d,e [DOI](#)). By 7 days post treatment, however, the basal-like cells that filled the SG were all positive for Lrig1 (Figure 4f,g [DOI](#) and Figure S4e), with the total number of Lrig1 positive cells per SG increasing significantly for aJ2 and aJ1J2 treated SGs (Figure 4h [DOI](#)), while SGs after aJ1 treatment showed normal morphology and marker expression (Figure S4c,d). The proportion of AR-expressing cells was not significantly different between treatments (Figure 4i [DOI](#)). These results indicate that the basal-like cells that fill the SG after blocking Notch signaling by aJ2 and aJ1J2 treatment, are stem/progenitor cells. Additionally, we examined the SGs for their proliferative capacity to confirm stem/progenitor function. In a normal SG, proliferation is restricted to the basal stem cells (Figure 4j [DOI](#)). As expected, proliferation was also restricted to the basal stem cells in aJ1 treated SGs (Figure S4f,g). Remarkably, most of the basal-like cells in the aJ2 treated SGs were proliferative, while the rare mature sebocytes remaining in the SG were non-cycling (Figure 4k [DOI](#)). As the SGs are mostly filled with basal-like cells at this time (Figure 4h [DOI](#)), the total number of proliferating cells per SG was significantly higher (Figure 4l [DOI](#)).

Overall, these data suggest that inhibition of Notch activity by aJ2 treatment retains the stem and progenitor (Lrig1+ and Lrig1+/AR+, respectively) cells in their immature proliferative state and prevents differentiation.

The block in sebocyte differentiation is lifted upon recovery of Notch activity

The therapeutic antibodies employed do not inhibit Notch signaling permanently, as the antibodies eventually become cleared from the animal's system. To determine whether the loss of sebocyte phenotype was reversible, we examined the SGs 14 days post single-dose treatment. Intriguingly, mature sebocytes begin to recover at this time point (Figure 5a,b [DOI](#), compare with Figure 1f,g [DOI](#)). We hypothesized that the sebocyte recovery must be due to the return of Notch activity after antibody washout. To test this, we examined the SGs for Notch activity at 7 days post treatment, since the return of Notch activity must precede the recovery of mature sebocytes. Fittingly, we observed N1ICD expression return in some of the basal-like cells at this time (Figure 5c,d [DOI](#) and Figure S5a-d). The percent of N1ICD positive cells per SG strongly increased from 1% at 3 days post treatment to 29% at 7 days post treatment (Figure 5e [DOI](#)), even though the percentage of N1ICD positive cells per SG remained significantly lower than control for aJ2 and aJ1J2 treated SGs at this time (Figure S5d). Next, we examined the SGs at 14 days post treatment for mature sebocyte markers to confirm the sebocyte recovery. Indeed, we saw the return of cells expressing adipophilin in the aJ2 and aJ1J2 treated SGs (Figure 5f,g [DOI](#) and Figure S5e-h). There was an overall

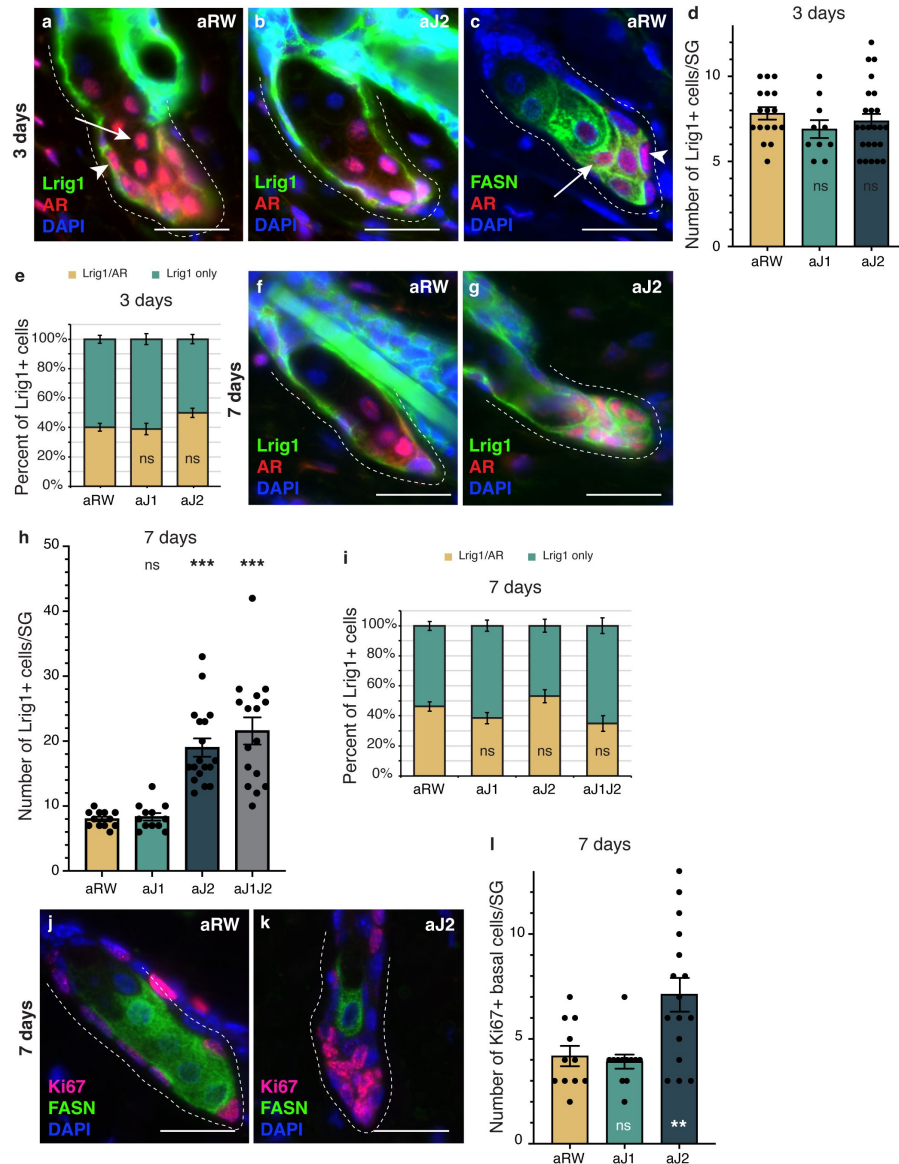


Figure 4.

Notch activity in the SG stem cells is required to prevent unregulated progenitor proliferation.

(a,b) Representative co-stain images for Lrig1 and AR in SGs from mice ($n = 5$ each) treated with aRW (a) and aJ2 (b), 3 days post treatment. (c) A representative co-stain image for FASN and AR in an SG from mice ($n = 5$ each) treated with aRW, 3 days post treatment. Arrows point to sebocytes, and arrowheads point to progenitor cells. (d) Quantification of the number of cells expressing Lrig1 in each SG, 3 days post treatment with aRW, aJ1 and aJ2. p-values: aJ1 = 0.152, aJ2 = 0.450. (e) Quantification of what percentage of the Lrig1+ cells express AR, 3 days post treatment with aRW, aJ1 and aJ2. p-values: aJ1 = 0.789, aJ2 = 0.028. (d and e) Total n of SGs quantified per treatment: aRW = 17, aJ1 = 10, aJ2 = 24. (f,g) Representative co-stain images for Lrig1 and AR in SGs from mice ($n = 5$ each) treated with aRW (f) and aJ2 (g), 7 days post treatment. (h) Quantification of the number of cells expressing Lrig1 in each SG, 7 days after treatment with aRW, aJ1, aJ2 and aJ1J2. p-values: aJ1 = 0.628, aJ2 = 9.06E-07, aJ1J2 = 7.88E-06. (i) Quantification of what percentage of the Lrig1+ cells express AR, 7 days after treatment with aRW, aJ1, aJ2 and aJ1J2. p-values: aJ1 = 0.117, aJ2 = 0.248, aJ1J2 = 0.091. (h and i) Total n of SGs quantified per treatment: aRW = 12, aJ1 = 12, aJ2 = 18, aJ1J2 = 16. (j,k) Representative co-stain images for Ki67 and FASN in SGs from mice ($n = 5$ each) treated with aRW (j) and aJ2 (k), 7 days post treatment. (l) Quantification of the number of cells expressing Ki67 in each SG, 7 days after treatment with aRW, aJ1, and aJ2. p-values: aJ1 = 0.652, aJ2 = 0.009. Total n of SGs quantified per treatment: aRW = 11, aJ1 = 12, aJ2 = 16. Student's T-test used for statistical analysis. All treatments compared against aRW. Error bars represent SEM. Scale bars are 25 μ m.

increase in the number of these cells from day 7 to day 14 (**Figure 5h**), even though the number of these cells remained significantly lower in these SGs compared to controls (**Figure S5h**). Interestingly, the majority of the adipophilin expressing cells were found in the proximal third of the SG (51% for aJ2 treatment and 66% for aJ1J2 treatment), consistent with the initiation of sebocyte differentiation at the proximal tip. However, there was a significant proportion of these cells found in the middle third (22% and 25% respectively), and distal third (27% and 9% respectively) of the SG. This could be due to the newly differentiated cells moving through the SG in a proximal to distal direction, as is the case during normal homeostasis. Alternatively, the stem cells could also be differentiating at sites other than just the proximal tip, as previously demonstrated by multi-color lineage tracing (Andersen et al., 2019). We further examined the SGs at 14 days post treatment for their AR expression, and confirmed that it had also been restored to its normal pattern (**Figure 5i,j** and **Figure S5i-l**). We also found that the average number of AR-expressing cells decreased from day 7 to day 14 post aJ2 and aJ1J2 treatment (10 to 2.95, and 8.56 to 5.55, respectively) (**Figure 5k**), returning to a more homeostatic state. Together, these data indicate that Notch inhibition does not result in a permanent cell fate switch, but maintains the stem/progenitor state, allowing the recovery of the differentiation process with the restoration of Notch activity.

Discussion

Based on our findings, we propose that the Jag2/Notch1 signaling axis is essential for correct sebocyte differentiation in homeostatic dorsal skin, and that inhibition of this signaling retains the basal stem and progenitor cells in a proliferative state, and blocks further differentiation. Thus, Notch signaling is required to prevent unregulated stem/progenitor proliferation, and induction of the sebocyte differentiation program. Moreover, Notch inhibition resulting in complete loss of mature sebocyte differentiation is a reversible phenotype, indicating that there remains a functional progenitor pool present during the studied timeframe.

Here, we have leveraged the use of monoclonal therapeutic antibodies designed to inhibit each of the distinct Notch receptors or ligands to study the role of Notch signaling in adult homeostatic tissue. We have shown that Jag2 is the hitherto unknown ligand involved in regulating sebocyte differentiation. Inhibition of Notch signaling using Jag2 blocking antibodies results in the loss of mature sebocytes, with the resulting SG being filled with basal-like cells forming “finger-like” epithelial remnants. A similar phenotype has been described by a previous study (Veniaminova et al., 2019) that used Lrig1-CreERT2 to irreversibly knock out RBPJ in adult homeostatic skin. This strategy enabled inhibition of Notch signaling specifically in the Lrig1⁺ stem cell population that normally maintains the SGs, but preserved it in the rest of the hair follicle. Interestingly, the authors saw two contradictory phenotypes as a result: overall loss of SG lobes, replaced by the finger-like epithelial remnants, as well as the presence of persistent SGs that were enlarged. The authors showed that over time, patches of Rbpj mutant cells extended out of their niche into the interfollicular epidermis (IFE), where Lrig1 is not expressed, and that loss of RBPJ in the IFE led to enlarged SGs. They argue that while Notch signaling promotes sebocyte differentiation in the SG stem cells, it indirectly suppresses these glands from the IFE. Our results did not show the enlarged SG phenotype, and we only observed the finger-like epithelial remnants. While injections of the Notch blocking antibodies are systemic, we only observed a reduction in the number of Notch-active cells in the IFE, but not a complete loss. This could explain why we didn’t observe the enlarged SG phenotype. Additionally, we observed the phenotype as early as 3 days post antibody treatment, while Veniaminova et al., observed the SGs 10 weeks after tamoxifen injection. These differences are likely due the different methodologies used in the two studies.

Several studies indicate that Notch signaling is essential for the postnatal maintenance of SGs (Blanpain et al., 2006; Estrach et al., 2008, 2006; Pan et al., 2004; Watt et al., 2008), but so far only Veniaminova et al., have specifically examined SGs in the adult homeostatic skin. They

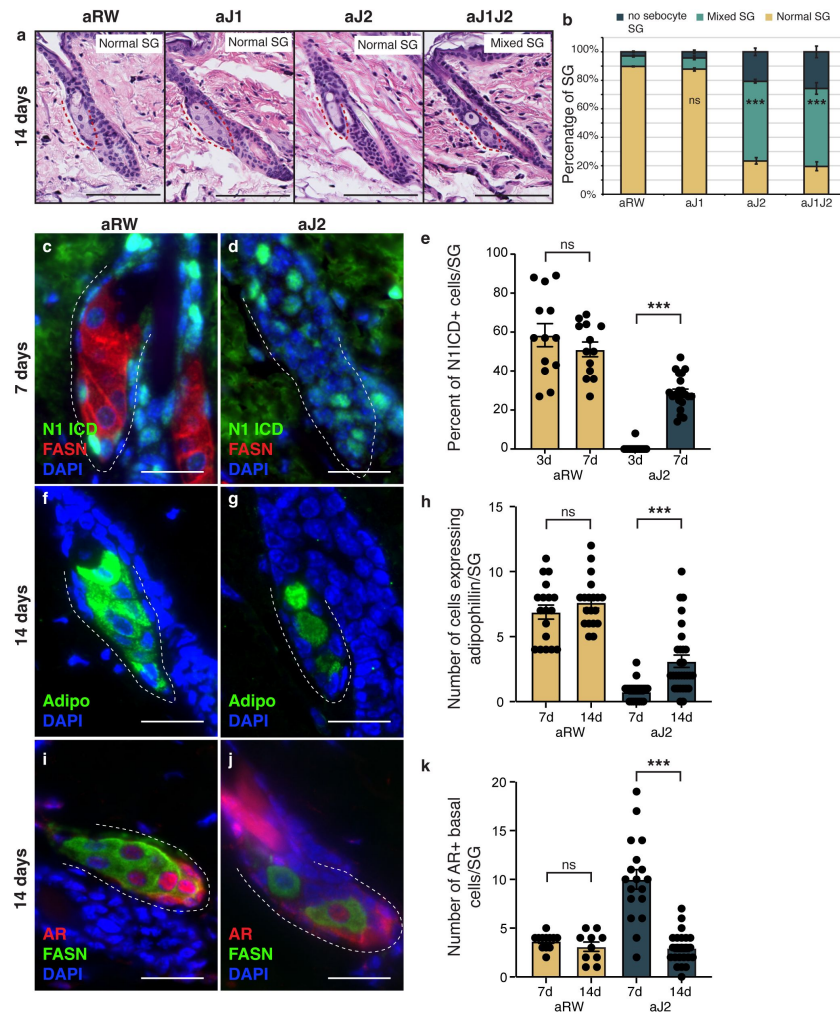


Figure 5.

The block in sebocyte differentiation is lifted upon recovery of Notch activity.

(a) Representative H&E images of SGs from mice ($n = 5$ each) treated with aRW, aj1, aj2 and aj1J2, 14 days post treatment. (b) Quantification of type of SG found after each treatment. The SGs were divided into three categories: normal SGs containing characteristic number of sebocytes (normal SG), SGs containing a mix of sebocytes and basal-like cells (mixed SG), and SGs containing no sebocytes and only basal-like cells (no sebocyte SG). p-values: aj1 = 0.535, aj2 = 2.90E-87, aj1J2 = 1.73E-92. Chi-square test used for statistical analysis. All treatments compared against aRW. Total n of SGs quantified per treatment: aRW = 485, aj1 = 430, aj2 = 394, aj1J2 = 364. Scale bars are 100 μm . (c,d) Representative co-stain images for N1ICD and FASN in SGs from mice ($n = 5$ each) treated with aRW (c) and aj2 (d), 7 days post treatment. (e) Quantification of the percentage of N1ICD+ cells in SGs from mice ($n = 5$ each) treated with aRW and aj2, 3 and 7 days post treatment. Percentage was calculated by dividing the number N1ICD+ cells by total number of basal-like cells in each SG. p-values: for comparison between day 3 and day 7 for aRW treatment = 0.297, for comparison between day 3 and day 7 for aj2 treatment = 1.00E-14. Total n of SGs quantified per treatment: aRW at 3days = 13, aRW at 7days = 13, aj2 at 3days = 15, aj2 at 7 days = 22. (f,g) Representative adipophilin staining in SGs from mice ($n = 5$ each) treated with aRW (f) and aj2 (g), 14 days post treatment. (h) Quantification of the number of cells expressing adipophilin in each SG, 7 and 14 days after treatment with aRW, and aj2. p-values: for comparison between day 7 and day 14 for aRW treatment = 0.292, for comparison between day 7 and day 14 for aj2 treatment = 1.38E-4. Total n of SGs quantified per treatment: aRW at 7days = 18, aRW at 14days = 19, aj2 at 7days = 22, aj2 at 14 days = 29. (i,j) Representative co-stain images for AR and FASN in SGs from mice ($n = 5$ each) treated with aRW (i) and aj2 (j), 14 days post treatment. (k) Quantification of the number of cells expressing AR in each SG, 7 and 14 days after treatment with aRW and aj2. p-values: for comparison between day 7 and day 14 for aRW treatment = 0.273, for comparison between day 7 and day 14 for aj2 treatment = 9.53E-09. Total n of SGs quantified per treatment: aRW at 7days = 12, aRW at 14days = 10, aj2 at 7days = 18, aj2 at 14 days = 23. Student's T-test used for statistical analysis. Error bars represent SEM. Scale bars are 25 μm .

observed that the stem/progenitor and differentiation markers were intermingled in the finger-like remnants forming an unnatural hybrid state, neither staying in a true progenitor state nor differentiating. In contrast, we observed a complete block in differentiation, with the stem/progenitor cells being locked in an immature proliferative state. These observations suggest that antagonistic antibodies may be able to achieve a more complete inhibition of Notch signaling in the SG stem/progenitor compartment.

Importantly, we were able to show that the loss-of-sebocyte phenotype in the SG is reversible. Having used therapeutic antibodies to block Notch signaling, the inhibition of the signaling pathway was not permanent. As Notch activity returned to the SG after antibody washout, sebocyte differentiation also recovered. These data indicate that the loss of Notch signaling does not impact the stem/progenitor potential of the SG, but prevents further differentiation. Indeed, a functional progenitor pool accumulates in the SG, primed for differentiation, as soon as Notch signaling becomes active. A recent study by Veniaminova et al., has shown that *Lrig1*-CreERT2 used to irreversibly knock out *Pparg* in adult homeostatic skin ablates 99% of the SGs (Veniaminova et al., 2023 [DOI](#)). Interestingly, they showed that non-recombined cells from other parts of the hair follicle migrate to the SG zone and regenerate the genetically ablated SGs. This regeneration process is dependent on the hair growth cycle, with the SGs primarily regenerating during anagen (active growth) phase and not in the telogen phase. While our studies cannot rule out the contribution of non-SG cells to the SG recovery seen upon return of Notch activity, this recovery is not dependent on the hair growth cycle. Sebocytes are able to differentiate while the hair follicle is still in the telogen phase, indicating a lift of the differentiation block on the stem/progenitor cells. This reversibility of phenotype is intriguing from a therapeutic perspective, as it suggests a translational potential in skin disorders involving sebocyte overactivity, such as acne.

An important open question is how Notch signaling regulates sebocyte differentiation. FASN, which can be used as a readout for sebocyte differentiation, is a downstream target of AR (Schirra et al., 2005 [DOI](#)). We observed an *Lrig1*⁺/AR⁺/FASN-population in the normal SGs, similar to the basal-like cells that fill the finger-like remnants seen after Notch inhibition. This expression pattern indicates that AR requires a co-activator to activate downstream gene expression in the SG. Previous studies have reported that Notch effectors such as HEY1, HEY2 and HEYL can act as co-repressors of AR in prostate cells (Belandia et al., 2005 [DOI](#); Kamińska et al., 2020 [DOI](#); Lavery et al., 2011 [DOI](#)). It is possible that certain Notch effectors can act as either co-activators or co-repressors of AR in a context-dependent manner. Further investigation will be needed to understand the exact molecular role Notch signaling plays in regulating sebocyte differentiation.

Methods

Animal strains and treatments

All animal experimentation was performed according to protocols approved by the Genentech Institutional Animal Care and Use Program (IACUC) committee. Female C57BL/6 mice were obtained from Charles River-Hollister, and were used for all experiments. The mice were 8 weeks old upon treatment. This time point was chosen to correspond to the resting (telogen) phase of the hair growth cycle, as SG size can vary by hair cycle stage. All mice were injected intraperitoneally with a single dose of blocking antibodies diluted in sterile saline at the following concentrations: anti-Jag1 at 20mg/kg, anti-Jag2 at 20mg/kg, anti-Jag1+anti-Jag2 at 20mg/kg+20mg/kg = 40mg/kg, anti-Notch1 at 5mg/kg, anti-Notch2 at 10mg/kg and anti-Notch1+anti-Notch2 at 5mg/kg+10mg/kg = 15mg/kg. Anti-Ragweed isotype control antibody was injected at concentrations to match the maximum dose of treatment antibodies.

Histopathological analysis and immunochemistry

Telogen mouse dorsal skin was collected at 3, 7 and 14 days post antibody injection. For frozen sections, skin samples were fixed in 4% paraformaldehyde in PBS, for 40 minutes at 4°C, washed, and then immersed in 30% sucrose in PBS overnight at 4°C. The tissue was then embedded in OCT (TissueTek), frozen immediately on dry ice, and stored at -80°C. Additional skin tissue from the same animals was also fixed with 10% neutral buffered formalin and then used to create paraffin-embedded sections. Hematoxylin and eosin (H&E) staining was performed by the Pathology core at Genentech.

The triple immunofluorescence stain for Notch1 ICD (IHC), and Notch1 mRNA (ISH) and Jag2 mRNA (ISH) was performed by the histopathology development group at Genentech. ACD LS 2.5 probes were ordered from Advanced Cell Diagnostics. RNAScope LS 2.5 Murine-Jag2_C1 (417518) nucleotides spanning from nt 552-1480 of reference sequence NM_010588.2, and Murine-Notch1_C2 (404648-C2) nucleotides spanning from nt 1153-1960 of reference sequence NM_008714.3. For positive control RNAScope LS 2.0 Murine-PPIB probe (313917) nucleotides spanning from nt 98~856 of reference sequence NM_011149.2 were used. For negative control RNAScope LS 2.0 DapB probe (312038) nucleotides spanning from nt 414~862 reference sequence EF191515 were used. For Immunohistochemistry, we used N1ICD (Cell Signaling, 4147), at 20 ug/ml working concentration.

The triple immunofluorescence for murine Jag2_Notch1 (dual ISH) with anti_Notch1 ICD (IHC) 3 Plex ISH_ISH_IHC in murine tissues using formalin-fixed, paraffin-embedded sections was performed on the Leica Bond-RX complete automation system using the RNAScope® LS Multiplex Reagent Kit (322800). Slides were baked and dewaxed on Leica Bond-RX and pretreated with Bond Epitope Retrieval Solution 2 (ER2) (AR9640) from Leica at 100°C for 40 minutes. After pretreatment, the probes were cocktailed, and hybridization was performed at 42°C for 120 minutes followed by amplification steps and developed with Opal 570 fluor at 1:1000 and Opal 690 fluor at 1:1500 from Akoya using the Opal Polaris 7 Auto Detection Kit (NEL811001KT). The immunohistochemistry was performed upon completion of ISH. The primary antibody incubation was 60 minutes at room temperature, followed by secondary antibody incubation for 30 min at room temperature with HRP conjugated Goat anti Rabbit (Perkin Elmer, NEF812001EA). The slides were then developed with Opal 780 fluor at 1:25 (Akoya, SKU FP1501001KT) following the manufacturer's instructions. Slides were imaged using Olympus VS200.

All other immunohistochemistry was performed using the following antibodies: N1ICD at 1:500 (Cell Signaling, 4147), FASN at 1:100 (BD, 610963), Adipophillin at 1:500 (Fitzgerald, 20R-AP002), Ki67 at 1:100 (Thermo Fisher Scientific, SP6 RM-9106-SO), Lrig1 at 1:200 (R&D Systems AF3688-SP), and AR at 1:250 (Thermo Fisher Scientific, MA1-150 AN1-15). Slides were imaged using the Leica Thunder microscope.

Imaging parameters were identical for all images. Images were processed using Fiji and Adobe Illustrator.

Acknowledgements

We are grateful to C. Cottonham, S. Hankeova, G. Hernandez for helpful discussions. We thank the Genentech Research Pathology, Necropsy, and Histology laboratories for their experimental contributions. We appreciate the insightful feedback and comments on the paper from L. Mosteiro, E. Reyes and B. Biehs.

Author contributions

S.N.F.A. designed the project, performed experiments and data analysis, and wrote the manuscript. S.C. developed the triple stain methodology, and performed the relevant staining. K.S. helped with some experimental design. The first observation of the phenotype came out of D.F.'s work. L.V. analyzed the data and contributed to writing the manuscript. F.P. provided histopathological analysis and discussion. C.W.S. directed and supervised the research, and analyzed the data. All authors approve the content of the manuscript.

Conflict of interest

All authors are or were employees of Genentech Inc. and/or hold shares in Roche.

C.W.S is currently employed at Gilead Sciences.

References

- Ables JL, Breunig JJ, Eisch AJ, Rakic P (2011) **Not(ch) just development: Notch signalling in the adult brain** *Nat Rev Neurosci* **12**:269–283 <https://doi.org/10.1038/nrn3024>
- Al-Zaid T, Vanderweil S, Zembowicz A, Lyle S (2011) **Sebaceous gland loss and inflammation in scarring alopecia: A potential role in pathogenesis** *J Am Acad Dermatol* **65**:597–603 <https://doi.org/10.1016/j.jaad.2010.09.774>
- Andersen MS *et al.* (2019) **Tracing the cellular dynamics of sebaceous gland development in normal and perturbed states** *Nat Cell Biol* **21**:924–932 <https://doi.org/10.1038/s41556-019-0362-x>
- Belandia B, Powell SM, García-Pedrero JM, Walker MM, Bevan CL, Parker MG (2005) **Hey1, a Mediator of Notch Signaling, Is an Androgen Receptor Corepressor** *Mol Cell Biol* **25**:1425–1436 <https://doi.org/10.1128/mcb.25.4.1425-1436.2005>
- Binczek E, Jenke B, Holz B, Günter RH, Thevis M, Stoffel W (2007) **Obesity resistance of the stearoyl-CoA desaturase-deficient (scd1 -/-) mouse results from disruption of the epidermal lipid barrier and adaptive thermoregulation** *bchm* **388**:405–418 <https://doi.org/10.1515/bc.2007.046>
- Blanpain C, Lowry WE, Pasolli HA, Fuchs E (2006) **Canonical notch signaling functions as a commitment switch in the epidermal lineage** *Genes Dev* **20**:3022–3035 <https://doi.org/10.1101/gad.1477606>
- Buitrago W, Joseph AK (2008) **Sebaceous carcinoma: the great masquerader** *Dermatol Ther* **21**:459–466 <https://doi.org/10.1111/j.1529-8019.2008.00247.x>
- Cottle DL, Kretschmar K, Schweiger PJ, Quist SR, Gollnick HP, Natsuga K, Aoyagi S, Watt FM (2013) **c-MYC-Induced Sebaceous Gland Differentiation Is Controlled by an Androgen Receptor/p53 Axis** *Cell Rep* **3**:427–441 <https://doi.org/10.1016/j.celrep.2013.01.013>
- Estrach S, Ambler CA, Celso CLL, Hozumi K, Watt FM (2006) **Jagged 1 is a β -catenin target gene required for ectopic hair follicle formation in adult epidermis** *Development* **133**:4427–4438 <https://doi.org/10.1242/dev.02644>
- Estrach S, Cordes R, Hozumi K, Gossler A, Watt FM (2008) **Role of the Notch Ligand Delta1 in Embryonic and Adult Mouse Epidermis** *J Invest Dermatol* **128**:825–832 <https://doi.org/10.1038/sj.jid.5701113>
- Frances D, Niemann C (2012) **Stem cell dynamics in sebaceous gland morphogenesis in mouse skin** *Dev Biol* **363**:138–146 <https://doi.org/10.1016/j.ydbio.2011.12.028>
- Jung Y, Tam J, Jalian HR, Anderson RR, Evans CL (2015) **Longitudinal, 3D In Vivo Imaging of Sebaceous Glands by Coherent Anti-Stokes Raman Scattering Microscopy: Normal Function and Response to Cryotherapy** *J Invest Dermatol* **135**:39–44 <https://doi.org/10.1038/jid.2014.293>

- Kamińska A, Pardyak L, Marek S, Wróbel K, Kotula-Balak M, Bilińska B, Hejmej A. (2020) **Notch signaling regulates nuclear androgen receptor AR and membrane androgen receptor ZIP9 in mouse Sertoli cells** *Andrology* **8**:457–472 <https://doi.org/10.1111/andr.12691>
- Karnik P, Tekeste Z, McCormick TS, Gilliam AC, Price VH, Cooper KD, Mirmirani P (2009) **Hair Follicle Stem Cell-Specific PPARγ Deletion Causes Scarring Alopecia** *J Invest Dermatol* **129**:1243–1257 <https://doi.org/10.1038/jid.2008.369>
- Kretschmar K, Cottle DL, Donati G, Chiang M-F, Quist SR, Gollnick HP, Natsuga K, Lin K-I, Watt FM (2014) **BLIMP1 Is Required for Postnatal Epidermal Homeostasis but Does Not Define a Sebaceous Gland Progenitor under Steady-State Conditions** *Stem Cell Rep* **3**:620–633 <https://doi.org/10.1016/j.stemcr.2014.08.007>
- Lafkas D *et al.* (2015) **Therapeutic antibodies reveal Notch control of transdifferentiation in the adult lung** *Nature* **528**:127–131 <https://doi.org/10.1038/nature15715>
- Lavery DN, Villaronga MA, Walker MM, Patel A, Belandia B, Bevan CL (2011) **Repression of Androgen Receptor Activity by HEYL, a Third Member of the Hairy/Enhancer-of-split-related Family of Notch Effectors*** *J Biol Chem* **286**:17796–17808 <https://doi.org/10.1074/jbc.M110.198655>
- Lovási M, Szegedi A, Zouboulis CC, Törőcsik D (2017) **Sebaceous-immunobiology is orchestrated by sebum lipids** *Derm-Endocrinol* **9**:0–0 <https://doi.org/10.1080/19381980.2017.1375636>
- Mosteiro L *et al.* (2023) **Notch signaling in thyrocytes is essential for adult thyroid function and mammalian homeostasis** *Nat Metab* **5**:2094–2110 <https://doi.org/10.1038/s42255-023-00937-1>
- Nelson BR, Hamlet KR, Gillard M, Railan D, Johnson TM (1995) **Sebaceous carcinoma** *J Am Acad Dermatol* **33**:1–15 [https://doi.org/10.1016/0190-9622\(95\)90001-2](https://doi.org/10.1016/0190-9622(95)90001-2)
- Niemann C, Horsley V (2012) **Development and homeostasis of the sebaceous gland** *Semin Cell Dev Biol* **23**:928–936 <https://doi.org/10.1016/j.semcdb.2012.08.010>
- Ostler DA, Prieto VG, Reed JA, Deavers MT, Lazar AJ, Ivan D (2010) **Adipophilin expression in sebaceous tumors and other cutaneous lesions with clear cell histology: an immunohistochemical study of 117 cases** *Mod Pathol* **23**:567–573 <https://doi.org/10.1038/modpathol.2010.1>
- Page ME, Lombard P, Ng F, Göttgens B, Jensen KB (2013) **The Epidermis Comprises Autonomous Compartments Maintained by Distinct Stem Cell Populations** *Cell Stem Cell* **13**:471–482 <https://doi.org/10.1016/j.stem.2013.07.010>
- Pan Y, Lin M-H, Tian X, Cheng H-T, Gridley T, Shen J, Kopan R (2004) **γ-Secretase Functions through Notch Signaling to Maintain Skin Appendages but Is Not Required for Their Patterning or Initial Morphogenesis** *Dev Cell* **7**:731–743 <https://doi.org/10.1016/j.devcel.2004.09.014>
- Rittié L, Tejasvi T, Harms PW, Xing X, Nair RP, Gudjonsson JE, Swindell WR, Elder JT (2016) **Sebaceous Gland Atrophy in Psoriasis: An Explanation for Psoriatic Alopecia?** *J Invest Dermatol* **136**:1792–1800 <https://doi.org/10.1016/j.jid.2016.05.113>

- Sato C, Zhao G, Ilagan MXG (2012) **An Overview of Notch Signaling in Adult Tissue Renewal and Maintenance** *Curr Alzheimer Res* **9**:227–240 <https://doi.org/10.2174/156720512799361600>
- Schirra F, Suzuki T, Richards SM, Jensen RV, Liu M, Lombardi MJ, Rowley P, Treister NS, Sullivan DA (2005) **Androgen Control of Gene Expression in the Mouse Meibomian Gland** *Investig Ophthalmology Vis Sci* **46** <https://doi.org/10.1167/iovs.05-0426>
- Schneider MR, Paus R (2010) **Sebocytes, multifaceted epithelial cells: Lipid production and holocrine secretion** *Int J Biochem Cell Biol* **42**:181–185 <https://doi.org/10.1016/j.biocel.2009.11.017>
- Seiffert K, Seltsmann H, Fritsch M, Zouboulis C (2007) **Inhibition of 5 α -Reductase Activity in SZ95 Sebocytes and HaCaT Keratinocytes In Vitro** *Horm Metab Res* **39**:141–148 <https://doi.org/10.1055/s-2007-961814>
- Shi VY, Leo M, Hassoun L, Chahal DS, Maibach HI, Sivamani RK (2015) **Role of sebaceous glands in inflammatory dermatoses** *J Am Acad Dermatol* **73**:856–863 <https://doi.org/10.1016/j.jaad.2015.08.015>
- Siebel C, Lendahl U (2017) **Notch Signaling in Development, Tissue Homeostasis, and Disease** *Physiol Rev* **97**:1235–1294 <https://doi.org/10.1152/physrev.00005.2017>
- Smith KR, Thiboutot DM (2008) **Thematic review series: Skin Lipids. Sebaceous gland lipids: friend or foe?** *J Lipid Res* **49**:271–281 <https://doi.org/10.1194/jlr.r700015-jlr200>
- Stenn KS, Sundberg JP, Sperling LC (1999) **Hair Follicle Biology, the Sebaceous Gland, and Scarring Alopecias** *Arch Dermatol* **135**:973–974 <https://doi.org/10.1001/archderm.135.8.973>
- Tran IT *et al.* (2013) **Blockade of individual Notch ligands and receptors controls graft-versus-host disease** *J Clin Invest* **123**:1590–1604 <https://doi.org/10.1172/jci65477>
- Veniaminova NA *et al.* (2019) **Niche-Specific Factors Dynamically Regulate Sebaceous Gland Stem Cells in the Skin** *Dev Cell* **51**:326–340 <https://doi.org/10.1016/j.devcel.2019.08.015>
- Veniaminova NA, Jia YY, Hartigan AM, Huyge TJ, Tsai S-Y, Grachtchouk M, Nakagawa S, Dlugosz AA, Atwood SX, Wong SY (2023) **Distinct mechanisms for sebaceous gland self-renewal and regeneration provide durability in response to injury** *Cell Rep* **42** <https://doi.org/10.1016/j.celrep.2023.113121>
- Watt FM, Estrach S, Ambler CA (2008) **Epidermal Notch signalling: differentiation, cancer and adhesion** *Curr Opin Cell Biol* **20**:171–179 <https://doi.org/10.1016/j.ceb.2008.01.010>
- Wilson A, Radtke F (2006) **Multiple functions of Notch signaling in self-renewing organs and cancer** *FEBS Lett* **580**:2860–2868 <https://doi.org/10.1016/j.febslet.2006.03.024>
- Wu Y *et al.* (2010) **Therapeutic antibody targeting of individual Notch receptors** *Nature* **464**:1052–1057 <https://doi.org/10.1038/nature08878>
- Yu J, Siebel CW, Schilling L, Canalis E (2020) **An antibody to Notch3 reverses the skeletal phenotype of lateral meningocele syndrome in male mice** *J Cell Physiol* **235**:210–220 <https://doi.org/10.1002/jcp.28960>

Editors

Reviewing Editor

Beate Lichtenberger

Medical University of Vienna, Vienna, Austria

Senior Editor

Lori Sussel

University of Colorado Anschutz Medical Campus, Aurora, United States of America

Reviewer #1 (Public review):

Summary:

In this study, Abidi and colleagues used Notch pathway neutralizing antibodies to inhibit sebaceous glands in the skin. The authors find that blocking either the Notch1 receptor or the Jag2 ligand caused loss of the glands and increased retention of sebaceous progenitor cells. Moreover, these glands began to reappear 14 days after treatment.

Strengths:

Overall, this study definitively identifies the Notch receptor/ligand combination that maintains these glands in the adult. The manuscript is clearly written and the figures are beautifully made.

In this resubmitted manuscript, the authors have adequately addressed all the previous critiques.

<https://doi.org/10.7554/eLife.98747.2.sa3>

Reviewer #2 (Public review):

Summary:

In this report Abidi et al. use an antibody against Jag2, a Notch1 ligand, to inhibit its activity in skin. A single dose of this treatment leads to an impairment of sebocyte differentiation and an accumulation of basal sebocytes. Consistently Notch1 activity, measured as cleaved form of the Notch1 intracellular domain, is detected in basal sebocytes together with the expression of Jag2. Interestingly the phenotype caused by the antibody treatment is reversible.

Strengths:

The quality of the histological data with a clear phenotype, together with the quantification represents a solid base for the authors claims.

This work identifies that the ligand Jag2 is the Notch1 ligand required for sebocyte differentiation.

From a therapeutic point of view, it is interesting that the treatment with the anti-Jag2 is reversible.

Weaknesses:

The authors use a single approach to support their claims.

Future in vitro studies will be needed to understand how Notch signaling induces sebocyte

differentiation (i.e. a cell-autonomous mechanism, a mechanism based on cell competition, etc.).

<https://doi.org/10.7554/eLife.98747.2.sa2>

Reviewer #3 (Public review):

Abidi et al. investigated the role of Notch signalling for sebaceous gland differentiation and sebocyte progenitor proliferation in adult mouse skin. By injecting antagonising antibodies against different Notch receptors and ligands into mice, the authors identified that the Notch1 receptor and, to a lesser extent, Notch2 receptor, as well as the Notch ligand Jagged2, contribute to the regulation of sebaceous gland differentiation. In situ hybridisation confirmed that treatment with anti-Jagged2 dramatically reduced the number of basal sebocytes staining for the transcriptionally active intracellular domain of Notch1. Loss of Notch activity in sebocyte progenitors robustly inhibited sebaceous gland differentiation. Under these conditions, the number of sebocyte progenitors marked by Lrig1 was not affected, while the number of proliferating basal sebocytes was increased. Upon recovery of Notch activity, sebaceous gland differentiation could likewise be recovered. By suggesting that Notch activity in sebocyte progenitors is required to balance proliferation and differentiation, these data bring valuable new and relevant findings for the skin field on the sebaceous gland homeostasis.

<https://doi.org/10.7554/eLife.98747.2.sa1>

Author response

The following is the authors' response to the original reviews.

We thank the editors and reviewers for their thoughtful comments on our manuscript. We greatly appreciated the suggestions and recommendations that helped us to improve the study. With adaptations, and inclusion of novel data and analyses, we have addressed all points raised, and hope that by these improvements the study further meets the standards for eLife.

Reviewer #1 (Recommendations For The Authors):

Minor text edits should be made.

(1.1) As a recent study from the Wong lab also showed sebaceous gland regeneration following complete ablation (Veniaminova et al., 2023), this finding should be mentioned in the text, and the abstract ("Most strikingly...") should be toned down.

We thank the reviewer for the positive feedback, and for highlighting this part of the study from the Wong lab. Although we cited this study in a different context, we had not discussed the sebaceous gland regeneration finding. We have now added this to the discussion section of the manuscript.

(1.2) Introduction: In lines 31-33 discussing the connection of sebaceous glands with skin disorders, the 5 references cited seem to replicate the citations from a similar sentence in Veniaminova et al., 2019. The authors should vary their citations, as there are likely other publications that can be cited here.

Additional references have been added.

Reviewer #2 (Recommendations For The Authors):

The manuscript is well written and the data are well presented in the figures.

We thank the reviewer for the positive feedback.

(2.1) Here are some points that could be taken into consideration to improve the manuscript:

- Row 75 "the primary" regulator could be changed to "a crucial".

We appreciate this suggestion and have made the text edit.

- Row 86 could be added: ...is the dominant ligand of the Notch signalling.

We have made the text edit as suggested.

(2.2) Row 107-109 from the quantification of Figure 1G and Figure 2 it seems that only the aj2 treatment has an SG phenotype. Why aj1 doesn't have any effect? (same is true in other figures). If the data on aj1 are maintained in the manuscript, this should be argued in the discussion section.

The reviewer is correct in noting that the aj1 treatment does not cause the phenotype, and this is indeed one of the key findings of the study. This is maintained throughout the manuscript. We have also cited references showing that embryonic and adult deletions of *Jag1* do not cause any sebaceous gland defects. All these data argue that *Jag1* is not the relevant Notch signaling ligand in sebocyte differentiation. We have further clarified this in the manuscript.

(2.3) Related to Figure 3G. As the Lrig1 stem cells can go towards both the sebocyte differentiation, or the sebaceous duct differentiation, it would be interesting to evaluate if the differentiation impairment caused by the antibody treatment affects in a similar manner (or not) the sebaceous duct differentiation. This could be tested through immunofluorescence, selecting markers of sebaceous duct.

We thank the reviewer for this thoughtful question. We are unable to find any unique markers of the sebaceous ducts (that are not expressed in other parts of the sebaceous gland, especially sebocytes) in the literature, thus, any analysis of markers would be confounded by its change of expression due to the loss of sebocytes.

However, we have evaluated the histology using bursting sebocytes releasing sebum as a proxy of a functional sebaceous duct. We have not found any significant differences between treatments using this metric (Fig. S1).

(2.4) As the word "therapeutic" is often underlined in the manuscript, maybe a few sentences on the transnational aspects of the results could be added to the discussion.

We thank the reviewer for highlighting this point. We have added this to the discussion.

(2.5) Figure 3 suggests that Jag2 is produced by basal sebocytes and used by these cells to induce sebocyte differentiation. I'm wondering if in an in vitro cell system (with a mixture of marked Jag2-expressing cells and marked Jag2-negative cells), it would be possible to understand if this mechanism of differentiation is a cell-autonomous mechanism or a mechanism based on cell competition (for instance, it would be possible

that the progenitors compete for their niche on the basal layer by pushing neighbouring basal cells to differentiate presenting them Jag2).

We thank the reviewer for the insightful suggestion. The mechanistic underpinning of how Notch signaling induces sebocyte differentiation is still unclear, and we find the reviewer's suggestion very interesting. However, establishing an *in vitro* model that captures the aspects mentioned, would require a lot of optimization and validation. To help rapid dissemination of our findings we elected to keep this out of the manuscript, but we will certainly consider it for future studies.

Reviewer #3 (Recommendations For The Authors):

(3.1) The authors focussed on mouse back skin sebaceous glands to analyse the phenotype. Are the effects also reproducible in the sebaceous glands of the mouse ears and tail epidermis? If so, the data should be strengthened by quantifying the phenotype using tail epidermal whole mounts (Braun et al., 2003; Development, PMID: 12954714), ideally by co-staining sebaceous glands for differentiation markers (e.g. FASN, Adipophilin) or lipid deposits (e.g., Oil red O). Also, the authors need to clarify how many sebaceous glands were scored per mouse. If not, please provide a rationale explaining the location restriction.

We thank the reviewer for pointing this out. Indeed, we have only incorporated data from the telogen dorsal skin of the animals. We have now more accurately reflected this in the revised manuscript. Additionally, we have added the number of sebaceous glands quantified in each figure per the reviewer's suggestion.

Since the stage of hair growth cycle can affect the sebaceous glands, we chose the resting (telogen) phase of the hair cycle to reliably study the sebaceous glands. At 8 weeks of age, hair follicles have uniformly entered the telogen phase. As subsequent re-entry into the anagen phase is asynchronous in the adult skin, the color of the dorsal skin of C57BL/6 mice can be used to determine whether the hair follicles are in the telogen phase or not. These reasons led us to choose this location, allowing us to study only telogen phase hair follicles.

We also point out that previously reported data (Estrach et al., 2006) did not show differences between dorsal and tail skin, so we assume the mechanisms must largely be conserved. However, as the reviewer rightfully points out, we cannot be sure and have, therefore, indicated the dorsal location throughout the manuscript.

(3.2) The micrographs in Figure 2 suggest that expression of both Jagged2 and Notch1 (intercellular domain) is not restricted to the sebaceous glands, as both molecules appear to be detected also in the isthmus and lower hair follicle. Of note, the online tool provided by the Kasper and Linnarsson labs (<http://linnarssonlab.org/epidermis/>) shows that both molecules are more widely expressed in mouse back skin. Please provide some analysis of the overall expression of these molecules in mouse skin. In line, is the observed effect of using the antagonising antibodies restricted to the sebaceous glands? Please provide additional data on proliferation and differentiation in the interfollicular epidermis, hair follicle cycling, and other skin compartments. For instance, the data published in the cited paper by Lafkas et al. (2005) suggest a thickening of the dermal adipocyte layer upon Jagged2 inhibition using monoclonal therapeutic antibodies.

The reviewer is correct in noting that expression of both Jag2 and Notch1 is not restricted to the sebaceous gland. The Notch signaling pathway is a well-known regulator for epidermal differentiation, and members of the pathway are expressed in various locations of the skin, including the interfollicular epidermis and the hair follicle. The expression and function of Notch signaling in these locations has been reviewed in (Hsu et al., 2014; Nowell and Radtke,

2013; Watt et al., 2008). We have also added zoomed out images showing expression of Jag2 and Notch1 in the skin (Figure S2e,f).

The effect of the antagonizing antibodies is not restricted to sebaceous glands, as we already noted in our discussion section: “While injections of the Notch blocking antibodies are systemic, we only observed a reduction in the number of Notch-active cells in the IFE, but not a complete loss.” The functional impact of the antibodies is likely beyond the sebaceous gland, as the reviewer points out, but understanding the full effect in other compartments, we consider beyond the scope of the current study.

In our previous study (Lafkas et al., 2015), the skin was examined at different animal ages/gender and using different antibody dosing regimens, which is the likely explanation for the differences observed. We have now quantified the width of the adipocyte layer and the IFE and show that there are no significant differences between treatments (Figure S1g-j). This together with the histology suggest that there are no significant differences in the differentiation and proliferation of these compartments.

(3.3) Since Jagged1 is a Wnt/beta-catenin target gene that is essential for (ectopic) hair follicle formation and differentiation (Estrach et al., 2006, Development, PMID: 17035290) and the sebaceous gland is widely considered as an epidermal compartment with absent/low Wnt/beta-catenin pathway activity during normal homeostasis (Lim & Nusse, 2013, Cold Spring Harbor Perspectives in Biology, PMID: 23209129), how is the expression of Notch1 and Jagged2 regulated upstream in sebocyte progenitors? It would be important to bring some more mechanistic insights into the upstream regulation of Notch activity. In line with comment 2, how are the compartment-specific effects molecularly regulated if the effects are not restricted to the sebaceous glands?

The reviewer is correct in noting that the Wnt pathway does not seem to be a likely candidate for driving sebocyte differentiation through Notch signaling. Indeed, Wnt inhibition is required for sebocyte differentiation (Merrill et al., 2001; Niemann et al., 2002), and the *Jag2* promoter region also does not contain TCF binding sites (Katoh and Katoh, 2006).

We speculate that Myc might regulate Notch signaling in the sebaceous gland. It is expressed in the sebaceous gland basal stem cells and has been reported to positively regulate sebocyte differentiation (Cottle et al., 2013). In addition, studies have shown that *Jag2* is a Myc target gene (Fiaschetti et al., 2014; Yustein et al., 2010). However, evaluating which upstream pathway potentially regulates Notch signaling, and resolving the regulatory network of sebocyte differentiation beyond the direct Notch ligands and receptors would require extensive *in vivo* modeling using KO and transgenic animals, which we consider to be beyond the scope of the current manuscript.

References

- Cottle DL, Kretschmar K, Schweiger PJ, Quist SR, Gollnick HP, Natsuga K, Aoyagi S, Watt FM. 2013. c-MYC-Induced Sebaceous Gland Differentiation Is Controlled by an Androgen Receptor/p53 Axis. *Cell Rep* 3:427–441. doi:10.1016/j.celrep.2013.01.013
- Estrach S, Ambler CA, Celso CLL, Hozumi K, Watt FM. 2006. Jagged 1 is a β -catenin target gene required for ectopic hair follicle formation in adult epidermis. *Development* 133:4427–4438. doi:10.1242/dev.02644
- Fiaschetti G, Schroeder C, Castelletti D, Arcaro A, Westermann F, Baumgartner M, Shalaby T, Grotzer MA. 2014. NOTCH ligands JAG1 and JAG2 as critical pro-survival factors in childhood medulloblastoma. *Acta Neuropathol Commun* 2:39. doi:10.1186/2051-5960-2-39
- Hsu Y-C, Li L, Fuchs E. 2014. Emerging interactions between skin stem cells and their niches. *Nat Med* 20:847–856. doi:10.1038/nm.3643

- Katoh Masuko, Katoh Masaru. 2006. Notch ligand, JAG1, is evolutionarily conserved target of canonical WNT signaling pathway in progenitor cells. *Int J Mol Med*. doi:10.3892/ijmm.17.4.681
- Lafkas D, Shelton A, Chiu C, Boenig G de L, Chen Y, Stawicki SS, Siltanen C, Reichelt M, Zhou M, Wu X, Eastham-Anderson J, Moore H, Roose-Girma M, Chinn Y, Hang JQ, Warming S, Egen J, Lee WP, Austin C, Wu Y, Payandeh J, Lowe JB, Siebel CW. 2015. Therapeutic antibodies reveal Notch control of transdifferentiation in the adult lung. *Nature* 528:127–131. doi:10.1038/nature15715
- Merrill BJ, Gat U, DasGupta R, Fuchs E. 2001. Tcf3 and Lef1 regulate lineage differentiation of multipotent stem cells in skin. *Genes Dev* 15:1688–1705. doi:10.1101/gad.891401
- Niemann C, Owens DM, Hülsken J, Birchmeier W, Watt FM. 2002. Expression of Δ NLef1 in mouse epidermis results in differentiation of hair follicles into squamous epidermal cysts and formation of skin tumours. *Development* 129:95–109. doi:10.1242/dev.129.1.95
- Nowell C, Radtke F. 2013. Cutaneous Notch Signaling in Health and Disease. *Cold Spring Harb Perspect Med* 3:a017772. doi:10.1101/cshperspect.a017772
- Watt FM, Estrach S, Ambler CA. 2008. Epidermal Notch signalling: differentiation, cancer and adhesion. *Curr Opin Cell Biol* 20:171–179. doi:10.1016/j.ceb.2008.01.010
- Yustein JT, Liu Y-C, Gao P, Jie C, Le A, Vuica-Ross M, Chng WJ, Eberhart CG, Bergsagel PL, Dang CV. 2010. Induction of ectopic Myc target gene JAG2 augments hypoxic growth and tumorigenesis in a human B-cell model. *Proc Natl Acad Sci* 107:3534–3539. doi:10.1073/pnas.0901230107

<https://doi.org/10.7554/eLife.98747.2.sa0>